CSA3717 - Software Testing

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1) A payroll system produces incorrect salary calculations due to logical errors in code. Apply defect classification techniques to categorize these defects. Explain how understanding defect classes improves testing. Illustrate how such defects can be prevented in future releases. Suggest improvements in testing strategy ?

Defect classification in a payroll systems :-

Introduction :-

→ A payroll system is a software used to calculate employee salaries, allowances, deductions, bonuses, taxes and net pay.

→ Logical errors in the code may cause incorrect salary calculations, resulting in employee dissatisfaction and financial losses.

→ Defect classification is the process of grouping software defects based on their type, cause, severity, and impact.

→ it helps testers identify the root cause of errors and develop effective test cases.

Classification of payroll Defect :-

1, calculation defects :- Incorrect basic salary,

overtime, (or) bonus calculations.

2) Logic defects :- Incorrect conditions for salary increments [or] deductions.

3) Data defects :- Incorrect employee salary, tax rate, (or) working hours.

4) Interface defects :- Incorrect transfer of Salary data between payroll and tax modules.

5) Boundary defects :- Errors when salary, overtime, [or] tax values reach minimum [or] maximum limits.

6) Integration defects :- Incorrect results when payroll interacts with attendance [or] banking systems.

Classificiation improves Testing :-

* Identifies the root cause of defects.
* Helps prioritize critical salary calculation errors.
* Improves test case Coverage
* Reduces repeated errors and testing time
* Ensures accurate salary payments and employee trust

Prevention in future Releases :-

* Follow coding standards and conduct code reviews.
* Use unit testing for salary formulas.
* Apply automated regression testing after code changes.

* Validate employee data and Salary inputs.
* Use version control and Continous integration.
* perform independent Verification of payroll Calculation's.

Improvements in Testing strategy :-

* unit testing :- Test each salary calculation Seperately.
* Boundary value testing :- Test minimum and maximum salary values.
* Equivalence partiontioning :- Test valid and invalid groups.
* Integration testing :- Verify attendance, tax and Payroll modules together.
* Regression testing :- Ensure updates do not break existing calculations.
* User acceptance testing :- Confirm salary results with HR and payroll staff.

Overview :-

→ A payroll system calculates employee salaries, bounces, overtime, taxes and deductions. Logical errors Can Cause incorrect salary payments and reduce employee trust.

→ Defect classification helps identify and Correct these errors.

Root Cause analysis :-

→ Root Cause analysis identifies why a defect occured. for example, an incorrect mathematical operator may substract an allowance instead of adding it.

→ Reviewing the code and checking the salary formula helps locate the actual cause.

Impacts :-

* Incorrect employee salary payments.
* wrong tax and insurance deductions.
* Delays in monthly payroll processing.
* Employee dissatisfication and loss of trust.
* possible finanicial and legal compilance issues.

Role of Defect classification in Testing :-

→ classification helps testers select suitable test cases for each defect type.

→ it improves test coverage, identifies high risk errors, reduces debugging time and ensures that important salary calculations receive Sufficient attention.

Test case Design and Execution :-

→ Test cases should Cover normal, invalid and boundary inputs.

→ Testers compare actual outputs with expected results to identify defects.

Defect prevention Techniques:-

* follow coding standards
* Conduct peer code reviews.
* Validate employee salary data.
* Use unit testing for salary formulas.
* apply automated regression testing
* Maintain proper documentation and Version Control.

Software Quality Assurance methods:-

→ Quality assurance includes unit, integration, system, and user acceptance testing.

→ Independent Verification of salary calculations and automated test execution helps ensure accuracy before releasing the software.

Continous Testing and Improvement:-

→ After every software update, regression tests should be executed to ensure existing features still work correctly.

→ Defect reports should be analysed regularly to identify recurring errors and improve the development process.

Recommendations for future releases:-

* Automate salary calculation tests.
* Prioritize critical and high severity defects.
* use realistic employee payroll data in testing.

* Integrate testing into the development pipeline.
* monitor payroll results after deployment.
* Maintain a record of defects, and their solution's.

Conclusion :-

* Defect classification is essential for identifying, analysing, preventing payroll system errors.
* Combining systematic testing, root cause analysis, and continous improvement helps deliver accurate salary calculations, reliable software, and greater employee satisfaction.